FIGURE 1

Constructing and testing a synthesized proxy

Researchers define the focal microfoundation and the unit of analysis.

WORDS / CORPORATE TEXTS

1 Characterize the corpus

Use unsupervised topic discovery to build one topic catalogue across all firms' texts.

2 Decontextualize topics

Three LLM readers screen descriptors, then review shared passages independently. Two votes retain a subject-specific topic. Context alone is insufficient.

DEEDS / OBSERVED BEHAVIOR

Choose the behavioral proxy

Select an established behavioral measure of the same focal microfoundation.

Account for other determinants

Specify firm-level determinants of the behavior and suitable panel fixed effects.

3 Synthesize the proxies

Use Ridge or XGBoost to predict behavior from retained topics' density (emphasis) and sentiment (orientation), together with the specified determinants and fixed effects.

SYNTHESIZED PROXY

Predicted behavior

A fitted value for each firm-year.

DECOUPLING

Observed minus predicted behavior

Intent requires independent evidence.

4 Validate the synthesized proxy

Convergent validity

Does the proxy track both narrative and behavioral measures?

Incremental validity

Does it explain an external outcome beyond both measures considered together?